

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Models the motion of the particle by forming a second order differential equation. (must have correct terms but allow sign errors)	AO3.3	M1	$M \frac{d^2 x}{dt^2} = -4M \frac{dx}{dt} - 8Mx$ $\frac{d^2 x}{dt^2} + 4 \frac{dx}{dt} + 8x = 0$
	Obtains correct differential equation	AO1.1b	A1	$\lambda^2 + 4\lambda + 8 = 0$ $\lambda = -2 \pm 2i$
	Forms and solves auxiliary equation for 'their' D.E.	AO1.1a	M1	Complex roots $\Rightarrow$ General solution is of the form:
	States a correct form of the general solution for 'their' auxiliary solution. (ft only if both M1 marks have been awarded)	AO1.1b	A1F	$x = Ae^{-2t} \cos(2t + B)$ $\dot{x} = -2Ae^{-2t} \cos(2t + B)$ $-2Ae^{-2t} \sin(2t + B)$
	Uses initial conditions to find arbitrary constants for 'their' solution	AO1.1a	M1	$\dot{x}(0) = 0 \text{ so,}$ $-2A \cos(B) - 2A \sin(B) = 0$
	Obtains correct value for one of 'their' constants (ft only if all M1 marks have been awarded)	AO1.1b	A1F	$\Rightarrow \tan B = -1$ $\Rightarrow B = -\frac{\pi}{4}$
	Obtains correct value for both of 'their' constants (ft only if all M1 marks have been awarded)	AO1.1b	A1F	$x(0) = 1$ $\Rightarrow \frac{A}{\sqrt{2}} = 1$
	Uses 'their' model to describe the motion of the particle either as a written description or shown on a clearly labelled graph.	AO3.4	A1F	$\Rightarrow A = \sqrt{2}$ $x = \sqrt{2}e^{-2t} \cos\left(2t - \frac{\pi}{4}\right)$ $\therefore$ the particle oscillates about O, with period π seconds and decreasing amplitude.
(b)	Refines their DE model to account for altered resistive force by introducing a new	AO3.5c	B1	$\frac{d^2 x}{dt^2} + \alpha \frac{dx}{dt} + 8x = 0$

coefficient for $\frac{dx}{dt}$			Critical damping $\Rightarrow$
Uses or states condition for critical damping	AO1.2	B1	$\lambda^2 + \alpha\lambda + 8 = 0$ must have equal roots
Deduces value for coefficient of $\frac{dx}{dt}$	AO2.2a	R1	$\alpha^2 = 32$ $\alpha = 4\sqrt{2}$
States resistive force	AO3.4	B1	Resistive force should have magnitude $4\sqrt{2}Mv$
Total 12 marks			

Q2.

(a) $m\ddot{x} = -4mn^2x - 2mk\dot{x}$

$\ddot{x} + 2k\dot{x} + 4n^2x = 0$

AG

M1A1

A1

3

(b) (i) $k = n$

$p^2 + 2np + 4n^2 = 0$

$(p + n)^2 + 3n^2 = 0$

M1

$p = -n \pm n\sqrt{3}i$

A1

$x = e^{-nt} (A \cos \sqrt{3}nt + B \sin \sqrt{3}nt)$

ft provided in correct form

A1F

$t = 0, x = a \Rightarrow a = A$

A1

$\dot{x} = e^{-nt} (-\sqrt{3}nA \sin \sqrt{3}nt + \sqrt{3}nB \cos \sqrt{3}nt)$

$-ne^{-nt} (A \cos \sqrt{3}nt + B \sin \sqrt{3}nt)$

m1

$$t = 0, \dot{x} = 0 \Rightarrow 0 = \sqrt{3}nB - nA$$

$$B = \frac{a}{\sqrt{3}} \left(= \frac{\sqrt{3}a}{3} \right)$$

A1F

$$x = e^{-nt} \left(a \cos \sqrt{3}nt + \frac{\sqrt{3}a}{3} \sin \sqrt{3}nt \right)$$

AG

A1F

7

$$(ii) \quad x = 0, a \cos \sqrt{3}nt + \frac{\sqrt{3}}{3} a \sin \sqrt{3}nt = 0$$

Condone verification with $t > 0$

M1

$$(e^{-nt} \neq 0)$$

$$\tan \sqrt{3}nt = -\sqrt{3}$$

AG

M1A1

3

$$(c) \quad (i) \quad k = 2n$$

$$p^2 + 4np + 4n^2 = 0$$

M1

$$(p + 2n)^2 = 0$$

M1

$$x = e^{-2nt}(A + Bt)$$

A1

3

$$(ii) \quad \text{critical damping}$$

A1F

1

[17]

Q3.

$$(a) \quad (i) \quad m\ddot{x} = -\frac{\lambda x}{a}$$

M1

$$= -amn^2 \frac{x}{a}$$

$$\ddot{x} = -n^2 x \quad \text{SHM}$$

A1

2

(ii) $T = \frac{2\pi}{n}$

B1

1

(b) (i) $m\ddot{x} = -amn^2 \frac{x}{a} - mkv$
3 appropriate terms attempted
Signs consistent

M1

A1

$$\ddot{x} + k\dot{x} + n^2 x = 0$$

AG

A1

3

(ii) $p^2 + \frac{5n}{2}p + n^2 = 0$

$$\left(p + \frac{5n}{4}\right)^2 - \frac{9n^2}{16} = 0$$

M1

$$p + \frac{5n}{4} = \pm \frac{3n}{4}, \quad p = -2n, \quad p = -\frac{n}{2}$$

$$2p^2 + 5np + 2n^2 = 0$$

$$(2p + n)(p + 2n) = 0$$

$$p = -\frac{n}{2}, p = -2n$$

A1

$$x = Ae^{-2nt} + Be^{-\frac{n}{2}t}$$

M1

$$t = 0, x = 0: A + B = 0$$

$$t = 0, \dot{x} = U$$

$$\dot{x} = -2nAe^{-2nt} - \frac{n}{2}Be^{-\frac{n}{2}t}$$

m1

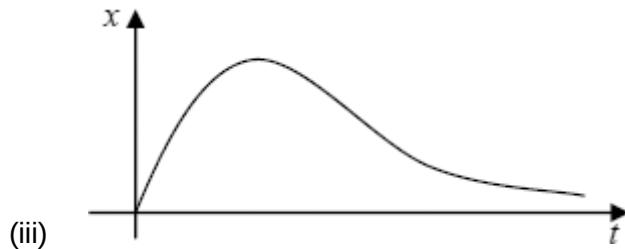
$$U = -2nA - \frac{n}{2}B$$

$$A = -\frac{2U}{3n} \quad B = \frac{2U}{3n}$$

$$x = \frac{2U}{3n} \left(e^{\frac{-nt}{2}} - e^{-2nt} \right)$$

A1,A1

6



Accept sketch with correct shape not reaching origin but not crossing x-axis elsewhere

Accept reference to real distinct roots of auxiliary equation

B1

Heavy damping

Independent of previous mark

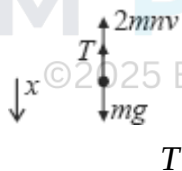
B1

2

[14]

Q4.

(a)



©2025 Exam Papers Practice. All rights reserved.

B1

$$m\ddot{x} = mg - \frac{5mn^2a}{a} \left(x + \frac{g}{5n^2} \right) - 2mn\dot{x}$$

all terms

M1A1A1

$$\ddot{x} + 2n\dot{x} + 5n^2x = 0$$

CAO; AG

A1

5

(b) AE: $p^2 + 2np + 5n^2 = 0$

M1

$$(p + n)^2 + 4n^2 = 0$$

$$p = -n \pm 2ni$$

A1

$$x = e^{-nt} (A \cos 2nt + B \sin 2nt)$$

A1

$$t = 0, x = 0 \Rightarrow A = 0$$

A1

$$\dot{x} = -e^{-nt} (B \sin 2nt) + e^{-nt} (2nB \cos 2nt)$$

M1

$$t = 0, x = U \Rightarrow$$

$$U = 2nB \quad B = \frac{U}{2n}$$

A1

$$x = \frac{U}{2n} e^{-nt} \sin 2nt$$

A1

7

EXAM PAPERS PRACTICE

(c) light

B1

©2025 Exam Papers Practice. All rights reserved.

1

(d) $x = 0 \therefore \sin 2nt = 0$

M1

$$2nt = (0, \dots) \pi, \dots$$

$$t = \frac{\pi}{2n}$$

A1

$$\dot{x} = -e^{-nt} \left(\frac{U}{2n} \sin 2nt \right) + e^{-nt} (U \cos 2nt)$$

$$= -e^{-\frac{\pi}{2}} (0) + e^{-\frac{\pi}{2}} (U)(-1)$$

M1

$$\text{speed} = Ue^{\frac{\pi}{2}}$$

A1

4

[17]

Q5.

- (a) Natural length of AP is $4a$ and natural length of BP is $2a$
When particle is x from equilibrium position:

$$\text{Tension in } AP \text{ is } \frac{4mn^2a(2a+x)}{4a}$$

M1A1

$$\text{Tension in } BP \text{ is } \frac{4mn^2a(a-x)}{2a}$$

M1A1

In general position, using $F = ma$:

$$m \frac{d^2x}{dt^2} = \frac{4mn^2(a-x)}{2a} - \frac{4mn^2a(2a+x)}{4a} - 2mn \frac{dx}{dt}$$

M1A1

$$m\ddot{x} = 2mn^2a - 2mn^2x - 2mn^2a - mn^2x - 2mn\dot{x}$$

$$\frac{d^2x}{dt^2} + 2n \frac{dx}{dt} + 3n^2x = 0$$

AG

©2025 Exam Papers Practice. All rights reserved.

A1

7

- (b) [Substituting $x = Ae^{pt}$]
 $p^2 + 2p + 3 = 0$

M1A1

$$p = -1 \pm \sqrt{2}i$$

A1

General solution is:

$$x = e^{-t}(A \cos \sqrt{2}t + B \sin \sqrt{2}t)$$

A1

$$\text{When } t = 0, \quad x = \frac{1}{2}a \Rightarrow \frac{1}{2}a = A$$

B1

Differentiating:

$$\frac{dx}{dt} = -e^{-t}(A \cos \sqrt{2}t + B \sin \sqrt{2}t) + e^{-t}(-A\sqrt{2} \sin \sqrt{2}t + B\sqrt{2} \cos \sqrt{2}t)$$

M1A1

$$\begin{aligned} \text{When } t = 0, \quad \frac{dx}{dt} &= 0 \\ \Rightarrow 0 &= -A + \sqrt{2}B \\ A &= \frac{1}{2}a, \quad B = \frac{1}{2\sqrt{2}}a \end{aligned}$$

A1

$$x = ae^{-t}\left(\frac{1}{2} \cos \sqrt{2}t + \frac{1}{2\sqrt{2}} \sin \sqrt{2}t\right)$$

8

[15]

Q6.

$$(a) \quad T_{AP} = \frac{\lambda \cdot 3a}{4a} = \frac{3}{4}\lambda$$

$$T_{PB} = 4mg \cdot \frac{a}{2a} = 2mg$$

Either

B1

Using $F = ma$ vertically

M1

$$mg + T_{PB} = T_{AP}$$

$$\therefore mg + 2mg = \frac{3}{4}\lambda$$

A1

$$\lambda = 4mg$$

A1

4

- (b) (i) When particle is moved a distance x below the equilibrium position, forces acting on it are

$$mg, T_{AP} = \frac{\lambda(3a+x)}{4a} = \frac{mg(3a+x)}{a},$$

M1

$$T_{PB} = 4mg \cdot \frac{(a-x)}{2a} = \frac{2mg}{a}(a-x)$$

and resistance $\frac{1}{5}mk\dot{x}$
All four forces

M1

[forces 2 and 4 are upwards]

Using $F = ma$ vertically downwards

$$m\ddot{x} = mg + T_{PB} - T_{AP} - \frac{1}{5}mk\dot{x}$$

Dependent on both M1 above

m1

$$m\ddot{x} = mg + \frac{2mg}{a}(a-x) - \frac{mg(3a+x)}{a} - \frac{1}{5}mk\dot{x}$$

A1

$$\ddot{x} - g - \frac{2g}{a}(a-x) + \frac{g(3a+x)}{a} + \frac{1}{5}k\dot{x} = 0$$

$$\ddot{x} + \frac{1}{5}k\dot{x} + \frac{3gx}{a} = 0$$

©2025 Exam Papers Practice. All rights reserved.

A1

$$10 \frac{d^2x}{dt^2} + 2k \frac{dx}{dt} + 5k^2x = 0$$

A1

6

(ii) Substituting $x = Ae^{nt}$,

$$10n^2 + 2kn + 5k^2 = 0$$

$$n = \frac{-2k \pm \sqrt{4k^2 - 200k^2}}{20}$$

M1

$$= \frac{1}{10}(-k \pm 7ki)$$

A1

$$x = e^{-\frac{k}{10}t} (A \cos \frac{7}{10}kt + B \sin \frac{7}{10}kt)$$

M1 A1ft

$$\text{When } t = 0, x = \frac{\alpha}{2}, A = \frac{\alpha}{2}$$

B1

Differentiating

$$\frac{dx}{dt} = -\frac{k}{10}e^{-\frac{k}{10}t} (A \cos \frac{7}{10}kt + B \sin \frac{7}{10}kt) +$$

$$e^{-\frac{k}{10}t} (-\frac{7}{10}kA \sin \frac{7}{10}kt + \frac{7}{10}kB \cos \frac{7}{10}kt)$$

M1 A1ft

$$\text{When } t = 0, \frac{dx}{dt} = 0, 0 = \frac{k}{10}A + \frac{7}{10}kB$$

M1

$$B = \frac{\alpha}{14}$$

$$x = \frac{\alpha}{14}e^{-\frac{k}{10}t} (7 \cos \frac{7}{10}kt + \sin \frac{7}{10}kt)$$

A1

©2025 Exam Papers Practice. All rights reserved.

9

[19]

Q7.

(a) Substituting $z = Ae^{nt}$,

$$2n^2 + 6n + 9 = 0$$

M1

$$n = \left(-6 \pm \frac{\sqrt{36 - 72}}{4} \right)$$

$$n = -\frac{3}{2} \pm \frac{3}{2}i$$

A1

$$\therefore x = e^{-\frac{3}{2}t} \left(A \cos \frac{3}{2}t + B \sin \frac{3}{2}t \right)$$

M1 A1ft

When $t = 0$, $x = a$; $a = A$

B1

Differentiating

$$\frac{dx}{dt} = -\frac{3}{2}e^{-\frac{3}{2}t} \left(A \cos \frac{3}{2}t + B \sin \frac{3}{2}t \right) + \frac{3}{2}e^{-\frac{3}{2}t} \left(-A \sin \frac{3}{2}t + B \cos \frac{3}{2}t \right)$$

Use of product formula

M1 A1

When $t = 0$, $\frac{dx}{dt} = 0$; $0 = -\frac{3}{2}A + \frac{3}{2}B$

M1

$$B = a$$

$$\therefore x = ae^{-\frac{3}{2}t} \left(\cos \frac{3}{2}t + \sin \frac{3}{2}t \right)$$

A1

9

- (b) Damping motion is light damping
And one reason

©2025 Exam Papers Practice. All rights reserved.

B1

since the particle oscillates, with its amplitude reducing to zero

B1

2

Or

$e^{-kt} \times \text{trig. term}$

Or

Sketch from graphics calculator

Full reason

[11]

Examiner reports

Q2.

This question was answered well in most cases. Part (a) was popular and mostly yielded full marks. Solutions to the auxiliary equation in part (b)(i) were often fully correct, with the most frequent error being the omission of 'i' in the solution, despite the obvious nature of the roots. Some chose a verification method for part (b)(ii) but only consideration of a general solution was sufficient for full credit. Solutions in part (c) were impressive.

Q3.

Candidates were able to attempt all parts of this question, but part (a) proved more challenging than expected, with some seeming unsure as to the requirements for simple harmonic motion. Part (b)(i) was done very well, as was part (b)(ii), although there were a number of errors in accuracy here. Part (b)(iii) proved very challenging; many successfully quoted the nature of the roots of the auxiliary equation as justification for heavy damping, but the hint of using a sketch was not appreciated.

Q4.

Part (a) was answered well, with occasional errors in finding the extension of the string. Part (b) was less successful, with some only being able to attempt the solving of the auxiliary equation before stopping. There were various algebraic errors, the most frequent being the omission of 'n' at varying stages, but still a number of good solutions. The answer to part (c) was well known. Part (d) proved quite testing in choosing correct methods for solution, and again algebraic errors marred solutions.

Q5.

Weaker candidates created the printed differential equation in part (a) by dubious means. It was necessary to find the equilibrium position and then use the extensions to be $2a + x$ in string AP and $a - x$ in string BP . Part (b) was answered very well by most candidates.

The only common error was in trying to insert the boundary conditions and finding $\frac{dx}{dt}$, when $\frac{d}{dx} \cos \sqrt{2}t$ sometimes became $-\sin \sqrt{2}t$ rather than $-\sqrt{2} \sin \sqrt{2}t$.

Q6.

Most candidates answered part (a) correctly but a number did not use $mg + T_{PB} = T_{AP}$ and fiddled the algebra to give the printed result, for which they gained no marks.

In part (b), candidates had to find tensions in the two springs AP and PB , with extensions $(3a + x)$ and $(a - x)$ respectively, and then use $F = ma$ downwards, giving

$m\ddot{x} = mg + T_{PB} - T_{AP} - \frac{1}{5}mkx$. Without these equations candidates were not given credit for producing the printed differential equation.

All candidates used the printed equation given in part (b)(i) to find x in part (ii). A number of candidates lost k in a variety of places, but virtually all obtained a result similar to that required.

Q7.

Part (a) was completed well by most candidates. However the use of the formula to solve the quadratic equation $2n^2 + 6n + 9 = 0$ was completed incorrectly by a number of candidates. Most appreciated that the damping was light, although the reason given for this was often imprecise.



EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.